nature machine intelligence

Perspective

https://doi.org/10.1038/s42256-025-01116-5

Towards deployment-centric multimodal AI beyond vision and language

Received: 28 March 2025

Accepted: 15 August 2025

Published online: 21 October 2025

Check for updates

A list of authors and their affiliations appears at the end of the paper

Multimodal artificial intelligence (AI) integrates diverse types of data via machine learning to improve understanding, prediction and decision-making across disciplines such as healthcare, science and engineering. However, most multimodal AI advances focus on models for vision and language data, and their deployability remains a key challenge. We advocate a deployment-centric workflow that incorporates deployment constraints early on to reduce the likelihood of undeployable solutions, complementing data-centric and model-centric approaches. We also emphasize deeper integration across multiple levels of multimodality through stakeholder engagement and interdisciplinary collaboration to broaden the research scope beyond vision and language. To facilitate this approach, we identify common multimodal-AI-specific challenges shared across disciplines and examine three real-world use cases: pandemic response, self-driving car design and climate change adaptation, drawing expertise from healthcare, social science, engineering, science, sustainability and finance. By fostering interdisciplinary dialogue and open research practices, our community can accelerate deployment-centric development for broad societal impact.

Data drives discovery in the twenty-first century[1]. From satellites monitoring climate change to social media capturing human behaviour, the variety and volume of information have never been greater. Each type of data, or modality, offers unique insights, but unimodal approaches often fall short of achieving robust or generalizable performance. Autonomous vehicles relying solely on visual data struggle with object detection in low-light or adverse weather conditions[2]. During the COVID-19 pandemic, relying solely on data from PCR with reverse transcription to diagnose SARS-CoV-2 infection led to frequent false negatives, hindering timely interventions[3]. Instead, integrating information from multiple modalities can reveal patterns and solutions that unimodal approaches miss[4].

Multimodal artificial intelligence (AI) leverages multimodal data to better understand complex systems through machine learning[5–8]. Its promise is evident in multidisciplinary applications such as pandemic response. For example, in healthcare, combining medical imaging, genomic sequencing and epidemiological data can enhance diagnoses, inform treatment strategies and support disease prevention[9–11]. In science, fusing genetic and protein structure data holds promise for advancing vaccine development[12]. In engineering, integrating textual specifications with spatial data can improve product design and manufacturing[13], which could extend to optimizing ventilator production. Across other disciplines, such as sustainability, finance and social science, multimodal AI offers the potential to provide deeper insights and actionable strategies[14–17].

While multimodal AI receives increasing attention and holds great promise, research has primarily focused on vision–language models[18,19] (Fig. 1 and Supplementary Fig. 1), leaving other modalities—such as tabular and time-series data—and related disciplines underexplored[20] (Fig. 2). Real-world challenges, such as pandemic response, call for AI capable of integrating diverse types of data through interdisciplinary collaboration, bridging the gap between research and application.

As modality integration broadens, multimodal-AI-specific deployment challenges increase, including missing modality[21,22], cross-modal alignment[8,23] and multimodal privacy risk[24,25]. These issues are compounded by broader barriers such as data limitations, integration

e-mail: h.lu@sheffield.ac.uk

https://doi.org/10.1038/s42256-025-01116-5

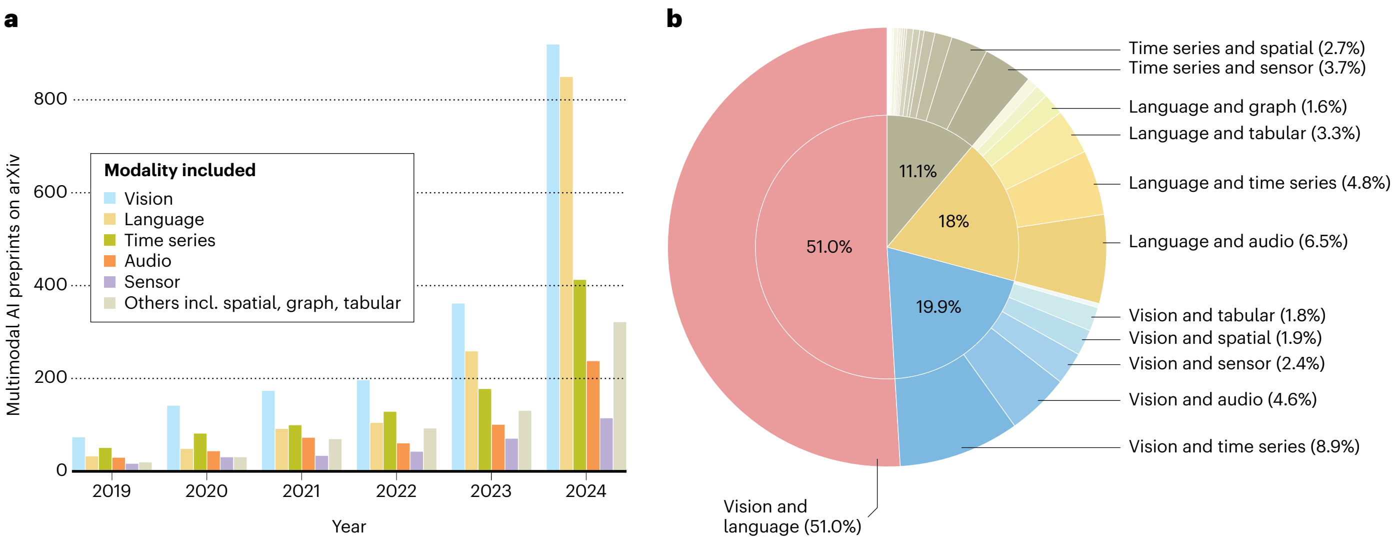

Fig. 1 | Trends in multimodal AI research (2019–2024) and the dominance of vision and language. a, Yearly growth of multimodal AI preprints on arXiv by modality, showing a steady increase over time and the dominance of vision and language. **b**, Breakdown of modality pairs in multimodal AI preprints on arXiv in 2024, revealing that over half the studies involve vision and language, followed by vision and others (19.9%), language and others (18.0%) and other modality combinations (11.1%). This analysis highlights the most common pairwise modality combinations and shows that those involving vision or language dominate, reaching 88.9%. For clarity and space efficiency, only modality pairs exceeding 1.5% are annotated. See Supplementary Information Section 2 for methodology and details of the trend analysis. See Supplementary Fig. 1 for a more detailed illustration and analysis of the multimodal AI landscape, including statistics on triple and quadruple modality combinations.

complexity and domain-specific constraints[26]. For example, rural healthcare may face limited computing infrastructure, financial services may face regulatory delays and real-time applications such as autonomous driving demand strict latency control. Multimodal set-ups often amplify these challenges due to increased system complexity. Simply combining diverse modalities is not enough; real-world success requires proactive alignment with deployment constraints from the outset.

This Perspective outlines a deployment-centric framework for multimodal AI that incorporates deployment constraints early on and addresses challenges specific to multimodal integration across disciplines. We first present a general workflow for developing deployment-ready multimodal AI systems. We then examine three data-intensive, cross-disciplinary use cases—pandemic response, self-driving car design and climate change adaptation—to illustrate common barriers and actionable strategies. By exploring underrepresented modalities and fostering open, interdisciplinary collaboration, we aim to broaden the impact of multimodal AI beyond vision and language.

Deployment-centric multimodal AI system development

Traditionally, multimodal AI research has been model-centric[5–7], focusing on developing new models to outperform existing ones on standardized benchmarks and datasets. The rise of generative AI[27] models such as ChatGPT[28,29] resulted in a shift towards data-centric approaches[30–32], emphasizing data resources and quality for better performance. However, the gap between high expectations and limited real-world impacts[26,33] indicates a pressing need for deployment-centric approaches that prioritize real-world applicability, user needs and ethical considerations, ensuring that AI innovations are both novel and practical, with a positive impact. We advocate a deployment-centric workflow for advancing multimodal AI from research to scalable solutions, built on ideas from machine learning operations lifecycle guidelines[34] and technology readiness levels for machine learning systems[35].

Figure 3a illustrates our deployment-centric workflow in three stages: planning, development and deployment. Planning focuses on defining the problem, determining the suitability of multimodal AI over unimodal AI, understanding real-world constraints and formulating AI tasks grounded in realistic assumptions. Development builds the multimodal AI system to learn predictive models from multimodal data, and deployment assesses real-world performance, feeding back insights to improve earlier steps. Guidance from stakeholder engagement and interdisciplinary collaboration (top right of Fig. 3a) informs the entire workflow, ensuring that the perspectives of domain experts, end-users and decision-makers are integrated for a robust and socially aligned system[36,37].

Planning for multimodal AI systems

Planning (the three blocks at the top left of Fig. 3a) begins with problem definition, which involves clearly articulating the problem's scope and objectives, and preliminarily assessing whether incorporating multimodal data can offer meaningful advantages, such as improved predictive performance or deeper insights, over unimodal approaches[38]. Beyond vision (image, video) and language (text), key modalities include audio, numeric time series (for example, financial data sequences), sensor signals (for example, wearable physiological measurements) and spatial (geolocation), tabular (structured data such as clinical records) and graph-based (relationships or networks) data. Furthermore, multimodality exists at multiple levels: multiple data types, subtypes, views and fidelities[39,40] (middle right of Fig. 3a, and Fig. 5). For example, in healthcare, multimodal data can range from different types—for example, medical images, electronic health records (EHRs) and omics data[41,42]—to subtypes within a data type (for example, X-ray, magnetic resonance imaging and ultrasound within the imaging modality), multiple views of the same data (sub)type (for example, anteroposterior, lateral and oblique views of X-rays) and multiple fidelities (for example, high, low and adaptive fidelities of anteroposterior X-rays). Under this broad definition, the potential number of modalities can expand substantially as more levels are considered. Understanding these modalities early on helps select those suited to the problem's complexity and capable of offering insights beyond unimodal data.

After establishing the potential benefits of multimodality, we move on to deployment constraints, which involves examining the AI

 https://doi.org/10.1038/s42256-025-01116-5

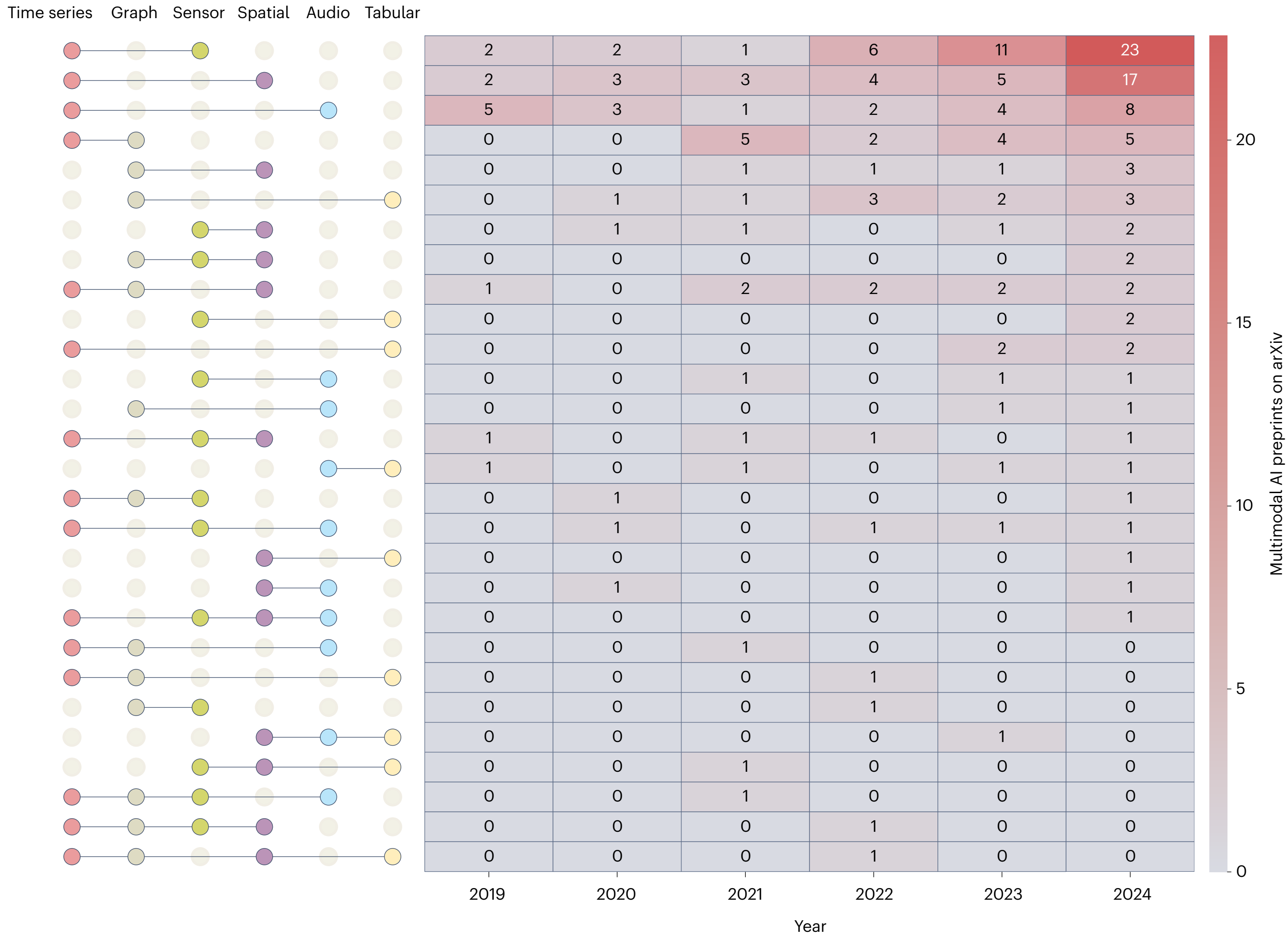

Fig. 2 | Underexplored modality combinations in multimodal AI (2019–2024). Heatmap of combinations of non-vision, non-language modalities in multimodal AI preprints by year. Each row represents a modality combination, and each column corresponds to a publication year. Darker shades indicate higher counts, with rows ordered by their 2024 totals. The coloured circles on the left identify the modality combinations. For example, in 2023, only two preprints used graph and tabular data (sixth row, fifth column). Time-series and sensor data form the most common combination, probably because sensor data are often recorded as time series. Time-series and spatial data are the second most common, possibly due to the importance of spatiotemporal modelling. By contrast, combinations involving graph, audio and tabular data remain sparsely studied. These gaps highlight untapped potential for multimodal AI beyond vision and language. See Supplementary Information Section 2 for details on data processing and modality extraction.

system's application context, including user needs, data availability, regulatory compliance[43], ethical considerations, societal impact and economic trade-offs between high-cost and low-cost data modalities. This step ensures that the chosen modalities and the resulting multimodal AI system are not only technically feasible but also viable within the intended deployment environments. Understanding these constraints early on informs task formulation, helping align solutions with real-world requirements and making it more effective and responsible.

The final step, task formulation, translates the defined problem and constraints into specific AI tasks, specifying the inputs, outputs and evaluation metrics required to meet the objectives set during problem definition. This provides a clear development roadmap, ensuring that multimodal AI systems are well positioned to meet both technical and practical needs. Selecting modalities requires balancing of utility, acquisition cost, complexity and deployment feasibility. Fewer well-curated modalities may outperform broader but less practical combinations.

Multimodal AI system development

Developing multimodal AI systems (lower left of Fig. 3a) parallels standard machine learning system development but introduces added complexities due to the integration of diverse data modalities. We have identified five multimodal-AI-specific challenges (Fig. 3b). Modality incompleteness occurs when one or more modalities are missing during training or deployment, necessitating the development of highly flexible or generative models. Multimodal heterogeneity deals with varying formats, scales and data structures, necessitating careful integration strategies to ensure interoperability. Cross-modality alignment ensures that the timing or meaning of data from different sources is correctly aligned for coherency and consistency, whether temporal (for example, matching timestamps) or semantic (for example, identifying data representing the same concept across modalities). Modality complementarity ensures that different modalities contribute complementary information to enhance performance, as more modalities may not improve results—for example, if they add noise or redundancy. Finally, multimodal privacy risk[24,25] arises when independently anonymized datasets become identifiable again (re-identification) through their fusion, thereby revealing sensitive information and necessitating robust privacy-preserving techniques.

The development process consists of five key steps: data collection, data curation, multimodal learning, evaluation and interpretation

 https://doi.org/10.1038/s42256-025-01116-5

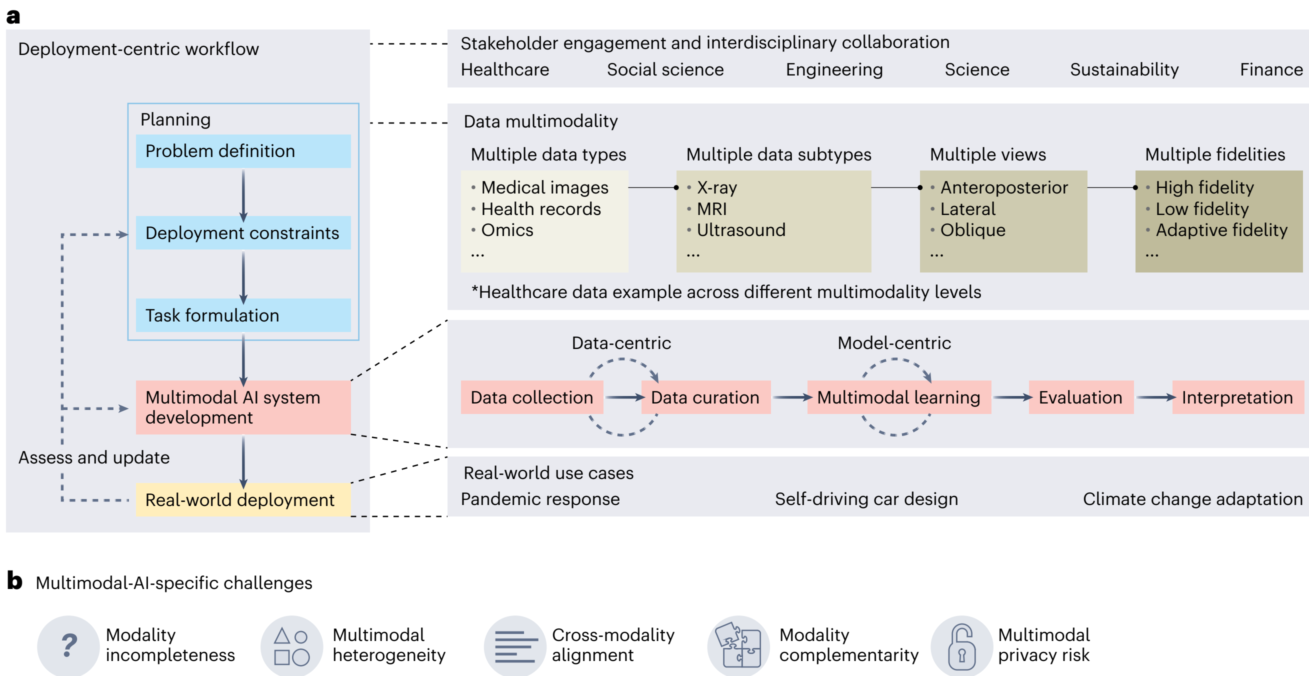

Fig. 3 | Deployment-centric multimodal AI: workflow and challenges. **a**, Deployment-centric multimodal AI workflow designed to meet real-world needs through a structured three-stage process covering planning, development and deployment, with iterative assessments and updates. In particular, the planning stage considers deployment constraints early on to ensure alignment with real-world settings and practical needs. This workflow incorporates stakeholder engagement and interdisciplinary collaboration at all stages to ensure that real-world needs and discipline-specific knowledge inform and enhance AI system development. Moreover, we consider multiple levels of data multimodality, from data types and subtypes to views and fidelities, as illustrated with a healthcare example. This broader definition of multimodality offers new perspectives and rich options for leveraging the benefits of multimodality. The system development stage has five steps, similar to a standard machine learning pipeline, where the data-centric and model-centric approaches[30] are indicated in the figure to highlight their differences from the deployment-centric approach. **b**, Five multimodal-AI-specific challenges shared across multiple disciplines and real-world applications: modality incompleteness, where one or more modalities are missing at training or deployment; multimodal heterogeneity, reflecting incompatible formats and data structures; cross-modality alignment, which requires synchronizing data in time or meaning; modality complementarity, where the goal is to maximize synergy without introducing redundancy; and multimodal privacy risk, where data fusion increases the chance of sensitive re-identification.

(lower right of Fig. 3a). Each step has a key role in addressing the multimodal-AI-specific challenges described above.

Data collection and curation build on the modality choices identified during planning, gathering relevant data from multiple sources and preparing them for AI model training. Diverse and reliable data sources ensure a high-quality, comprehensive representation of the problem space. Synthetic data and weak supervision provide valuable alternatives when labelled data are scarce[44], as in rare diseases or time-sensitive crises, where expert annotation may be infeasible. Once collected, the data undergo standard curation, such as wrangling, cleaning, annotation, handling missing values and quality assurance[26], tailored to multimodal AI. For instance, in healthcare, linking multimodal data from EHRs, imaging and biosignals poses substantial challenges due to privacy concerns[24] and the need for standardization. The integration and cross-referencing of these modalities can amplify the risk of re-identification, underscoring the importance of addressing multimodal privacy risk through robust privacy-preserving techniques[45–47]. Moreover, selected modalities should be interoperable and complementary, with their alignment and integration resulting in a cohesive, high-quality dataset suited to technical and deployment requirements.

Multimodal learning integrates diverse data modalities to leverage their unique strengths for improved predictive performance[5]. Traditional fusion strategies, early fusion (combining raw features), intermediate fusion (merging processed features) and late fusion (integrating outputs from independently processed modalities) offer trade-offs between model complexity and when modalities interact. Recent advances, hybrid fusion[4] and knowledge distillation[48], provide greater flexibility, enabling the combination of multiple strategies or the transfer of knowledge from complex to simpler models. Techniques such as co-attention mechanisms[49] can further enhance integration and performance by dynamically adapting to cross-modal interactions. The choice of strategy depends on the specific problem and data characteristics, balancing integration benefits with the added complexity of managing multiple modalities.

Evaluation and interpretation support the reliability, effectiveness and fairness of multimodal AI systems, especially in complex, high-stakes environments. Beyond standard performance metrics such as accuracy, evaluation should assess the contribution of individual modalities, the alignment and synergy between them, and the system's robustness to noise and variability. For example, evaluating safety in multimodal models[50] is key for understanding how biases across modalities may interact or amplify one another[51], potentially undermining system reliability. Interpretation[52,53] ensures that model decisions are understandable and transparent. Techniques such as heat maps, *t*-distributed stochastic neighbour embedding and decision trees help visualize and explain how different modalities interact and contribute to outcomes, fostering user trust and supporting error analysis and model refinement. Uncertainty estimation[54], via ensembling, calibration or Bayesian approaches, supports more reliable decisions in high-stakes applications.

Real-world deployment of multimodal AI systems

Deploying multimodal AI systems (bottom of Fig. 3a) requires infrastructure for hosting, scaling and ongoing management, with continuous monitoring to ensure that the system remains effective and

https://doi.org/10.1038/s42256-025-01116-5

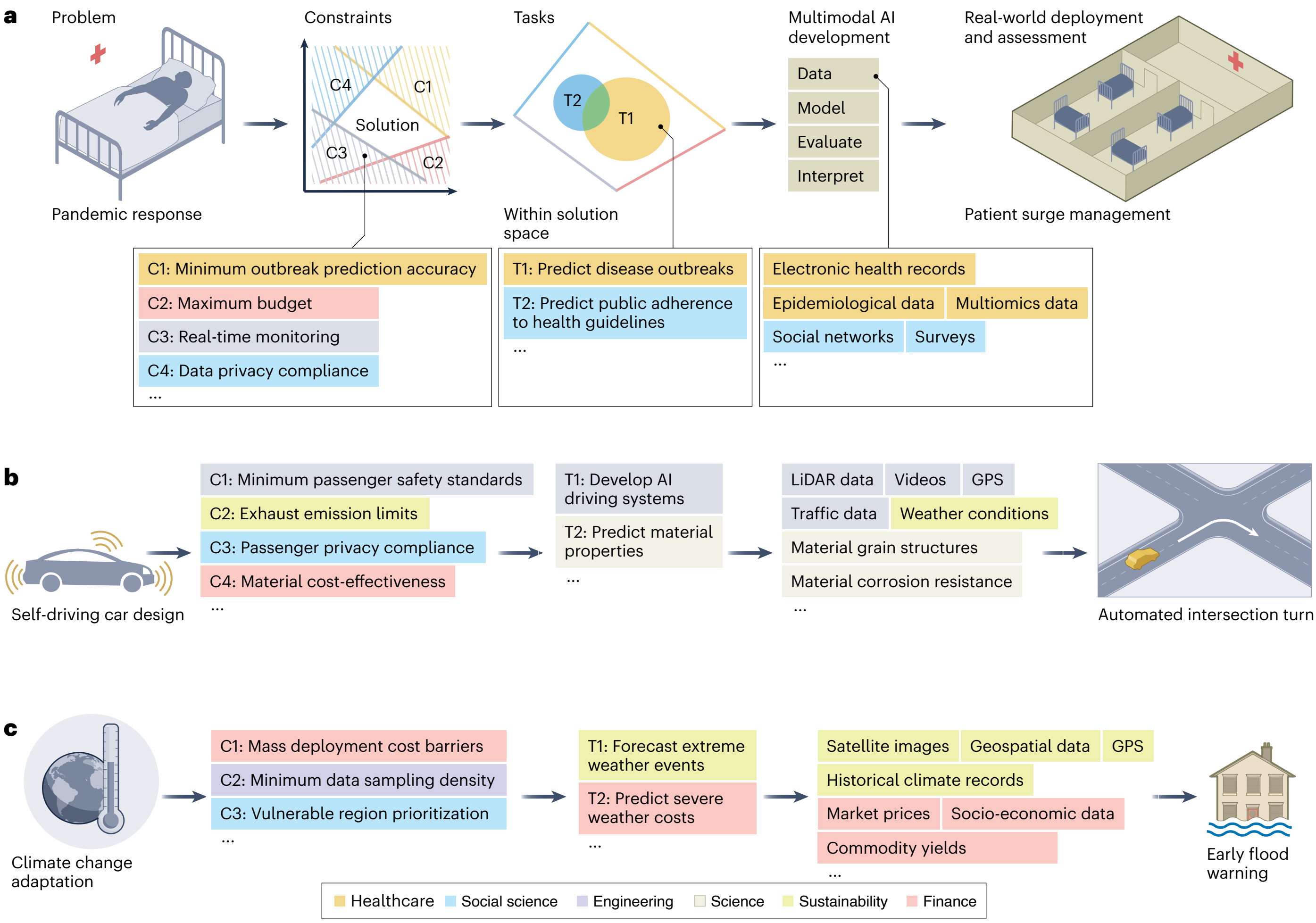

Fig. 4 | Deployment-centric multimodal AI workflow for three use cases. These examples demonstrate the versatility of the proposed workflow, showing how deployment-centric considerations can be applied across various disciplines to solve complex challenges. **a**, This example illustrates how the proposed deployment-centric multimodal AI workflow can better address real-world challenges in pandemic response, drawing on stakeholder engagement and interdisciplinary collaboration, as indicated by the different colours across stages. The process begins with defining the problem of interest properly and carefully considering the space of various deployment constraints to formulate specific tasks. Two tasks are shown in the figure: one with a focus on healthcare and the other on social science. Next, we develop multimodal AI systems using diverse data sources. Finally, we proceed to real-world deployment and assessment, exemplified by the application of patient surge management. **b**, In self-driving car design, the workflow incorporates deployment constraints from social science, engineering, sustainability and finance disciplines. Two key tasks focus on engineering and science considerations, respectively. Multimodal AI development uses multiple data sources, including remote sensing, weather and materials data, aiming for real-world deployment exemplified by automated intersection turn. **c**, In climate change adaptation, the workflow considers constraints such as mass deployment cost barriers, minimum data sampling density and prioritization of vulnerable regions. Task formulation and multimodal AI development focus on environmental sustainability and financial considerations, reflecting how task and model development are shaped by discipline-specific needs. Early flood warning serves as an example of real-world deployment and assessment.

relevant[26]. Infrastructural and regulatory conditions, including compute availability, connectivity and legal safeguards, can determine what is practically feasible and where deployment may succeed or fail. Unlike unimodal AI, these systems face unique challenges, such as integrating diverse data modalities in real-time environments and maintaining performance despite modality-specific issues, such as sensor failures or data stream interruptions. Robust monitoring mechanisms enable the system to detect and compensate for failures in one modality by leveraging information from others. Additionally, real-time multimodal data fusion, where multimodal data may arrive asynchronously, requires careful infrastructure design and adaptive algorithms.

The following sections examine three cross-disciplinary use cases that demonstrate the value and applicability of the deployment-centric multimodal AI workflow: pandemic response, self-driving car design and climate change adaptation.

Pandemic response

Pandemic response (Fig. 4a) presents a complex challenge requiring coordinated efforts across healthcare and other disciplines[55]. Defining the problem involves assessing early on how multimodal AI can offer advantages over unimodal approaches by integrating diverse data sources for deeper insights and more accurate predictions. For example, in social science, combining textual content from surveys or social networks with tabular demographic data can help us understand mental health impacts and detect changes in online behaviour[56,57]. In sustainability, analysing time-series environmental data, satellite imagery and sustainability reports can reveal the environmental effects of pandemics[58,59]. In finance, integrating economic indicators, transaction patterns and market response data with health data can predict socio-economic impacts and inform public health strategies[60,61]. A well-defined problem ensures that clear constraints and tasks can be established.

Datasets	Year of release	Data origin	Primary data modalities	Access	Scale	Example applications
TCGA	2008	US	Images, texts, omics	–	~11,000 individuals	Discovery of cancer genes and mutations
MIMIC-IV	2023	US	Texts, numerical data, time series	–	~231,000 individuals	Prediction of mortality in patients with lung cancer
Trafficking-10k	2017	US, CA	Images, texts	x	~10,000 advertisements	Human trafficking detection
*IGDD	2022	US	Texts (chat histories, surveys)	x	~26,700 conversations	Detection of adolescent online risks
nuScenes	2019	US, SG	Images, radar data, LiDAR data	–	~1,400,000 images	Prediction of vehicle traffic trajectories
DAIR-V2X	2021	CN	Images, LiDAR data	–	~71,300 LiDAR and camera frames	3D object detection
Materials Project	2013	Worldwide	Images, texts, graphs	–	~160,000 materials	Discovery of new inorganic crystals
MathVista	2024	Worldwide	Images, texts, symbolic data	–	~6,000 mathematical problems	Mathematical problem-solving
iNaturalist	2017	Worldwide	Images, timestamps, coordinates, species data	–	~118,000,000 observations	Species recognition
†ERA5	2018	Worldwide	Spatiotemporal data (temperature, humidity, wind)	–	~745,000 latitude–longitude grids	Reconstruction of historical climate extremes
Monetary policy calls	2022	Worldwide	Videos, texts, time series	–	~340 video conference calls	Prediction of gold price movements
China A-shares market	2022	CN	Texts, time series, graphs	–	~5,130,000 news articles	Forecasting of stock price movements

Healthcare; Social science; Engineering; Science; Sustainability; Finance; – Open access; x Restricted access

*Subtypes of text are considered as multiple modalities.
†Different 'views' of spatiotemporal data are considered as different modalities[46,47].

Fig. 5 | Examples of multimodal benchmark datasets. We selected two illustrative examples for each of the six disciplines to showcase diverse multimodal data and provide a starting point for multimodal AI exploration. For each dataset, we report six key attributes: year of release, data origin, data modalities, accessibility, scale of the primary modalities and example applications. In each discipline, datasets are listed in ascending order by year of release. For the healthcare, science and finance disciplines, we selected one small-scale dataset for quick experimentation and one larger-scale dataset for comprehensive exploration. The datasets include TCGA (The Cancer Genome Atlas)[114], MIMIC-IV (The Medical Information Mart for Intensive Care IV database)[115], Trafficking-10k (human trafficking advertisement dataset)[116], IGDD project (Instagram Data Donation project)[117], nuScenes (autonomous driving scene dataset)[118], DAIR-V2X (real-scenarios vehicle to everything dataset)[119], Materials Project (material property dataset)[120], MathVista (mathematical reasoning dataset)[121], iNaturalist (citizen science platform for biodiversity data)[122], ERA5 (European Centre for Medium-Range Weather Forecasts Reanalysis v.5)[123], Monetary policy calls (monetary policy call dataset)[124] and China A-shares market (dataset of public companies listed in China A-shares market)[125]. Modality definitions can vary. IGDD and ERA5 are not strictly multimodal in the traditional sense, but we treat them as such under the broader definition presented in this Perspective. The IGDD dataset contains only text data, but we consider its subtypes (chat histories and surveys) as distinct text modalities. Similarly, ERA5 comprises spatiotemporal variables such as temperature, humidity and wind, which we treat as complementary 'views' offering distinct information streams for climate modelling, as adopted in previous work[126,127].

The deployment of multimodal AI in pandemic response faces privacy, technical and economic constraints. Privacy concerns arise from integrating diverse data sources such as EHRs, social media and genomic information, which increases the risk of re-identification and amplifies the potential impact of data breaches. Maintaining public trust requires robust privacy-preserving techniques[45–47] and compliance with regulatory frameworks. Technical challenges include real-time monitoring, where multimodal AI systems must integrate and process data streams rapidly for timely interventions. The heterogeneity of health and social data complicates alignment and standardization, particularly in international or multicentre collaborations. Achieving minimum predictive accuracy thresholds for outbreak detection and public health decision-making helps build trust in AI-driven systems. Economic constraints, including computational costs and budget limitations, restrict accessibility and scalability. High-cost modalities (for example, magnetic resonance imaging) may be impractical in low-resource settings, so multimodal AI systems should flexibly support lower-cost alternatives (for example, wearables) to promote equitable deployment.

Task formulation translates the defined problem into specific AI objectives, detailing inputs, outputs and evaluation metrics for pandemic-related challenges. A primary task is predicting disease outbreaks by integrating multimodal data such as epidemiological records, EHRs and environmental factors to inform early interventions and resource allocation. Another task is monitoring public adherence to health guidelines using social media data, geospatial information and biosignals, providing actionable insights to refine policies. Additionally, multimodal AI can support real-time decision-making in healthcare by combining patient data streams, including wearable biosignals, medical images and clinical notes, to manage patient surges effectively[62,63]. Task formulation ensures that multimodal AI systems align with technical and practical needs, strengthening pandemic response through actionable insights and tailored interventions.

Developing multimodal AI systems for pandemic response follows the standard workflow of the five key steps outlined above. Data collection assembles diverse sources, such as EHRs, genomic data and biosignals, to create comprehensive datasets for outbreak prediction and patient management. During curation, standardization and alignment address challenges such as integrating time-sensitive data streams from biosignals and geospatial sources. Multimodal learning can use self-supervised[64] and cross-domain[65] techniques to capture cross-modal relationships effectively and integrate domain-specific knowledge. Evaluation requires multimetric, multicentre validation[66,67], ensuring robustness for tasks such as outbreak prediction and public adherence monitoring. Interpretability and explainability enable healthcare professionals and policy-makers to trust and act on multimodal AI insights in high-stakes scenarios[68]. Tailoring these development steps to pandemic-specific needs ensures adaptability, reliability and ethical compliance in rapidly changing contexts.

Deploying multimodal AI systems in pandemic response translates research into actionable solutions for high-stakes environments. For instance, patient surge management can integrate biosignals, medical images and clinical notes to optimize intensive care unit bed allocation and staff deployment. Public health decision-making can benefit from multimodal analyses of epidemiological data, EHRs and social media trends to guide interventions such as lockdowns or resource distribution. Real-time deployment requires robust infrastructure to process asynchronous data streams and address modality-specific interruptions, such as gaps in biosignal or social media data. Adaptive algorithms and reliable systems ensure timely and accurate outbreak prediction and monitoring. Assessment through multicentre trials and multidisciplinary benchmarks[69] can validate the added value of multimodal AI over unimodal systems. Measuring the accuracy of outbreak prediction or the effectiveness of intervention informed by multimodal AI can ensure trust and accountability. Further addressing scalability and privacy compliance can support sustainable and ethical deployment.

Self-driving car design

Self-driving car design (Fig. 4b) exemplifies a transformative application of multimodal AI, addressing challenges in safety, efficiency and sustainability[70,71]. Defining the problem involves determining how multimodal AI can substantially outperform unimodal approaches for navigating complex environments safely and efficiently. For instance, fusing sensors such as LiDAR (light detection and ranging), radar and cameras can enhance perception and decision-making in dynamic settings, especially where a single sensor may fail[72]. Beyond perception, multimodal AI can support physical simulations, such as combining text, images and videos, to model vehicle dynamics and road interactions[73]. In materials design, integrating structural and chemical data aids in developing lightweight, sustainable materials for vehicle construction[74]. Clearly defining these challenges ensures that multimodal AI solutions are effectively tailored to the demands of autonomous driving.

The deployment of multimodal AI in self-driving car design faces safety, privacy, technical and economic constraints that must be addressed for safe and scalable adoption. Privacy concerns arise from integrating multimodal data, such as LiDAR, cameras and passenger-related information, which increases the potential for data misuse or unauthorized tracking[75]. Mitigating these risks requires robust data anonymization and adherence to privacy regulations[76]. Technical challenges include real-time data processing in dynamic environments, such as urban intersections or adverse weather. Modality-specific failures, such as LiDAR disruptions in heavy rain, require fallback mechanisms, such as radar, to maintain reliability. Self-driving systems must achieve high predictive accuracy for collision avoidance and route planning, and operate reliably across regions with varying regulations and infrastructure. Economic constraints involve balancing the cost of high-resolution sensors with scalability needs. Meeting sustainability goals requires energy-efficient operation, with trade-offs between sensor performance and power consumption. Addressing these constraints holistically enables robust and accessible deployment of self-driving technologies.

In self-driving car design, task formulation maps problems and constraints to concrete AI objectives for perception, navigation and safety. A key task is developing AI driving systems that integrate multimodal data streams, including LiDAR, radar, Global Positioning System (GPS) and cameras, for real-time navigation. These systems must dynamically process inputs to handle traffic, obstacles and weather variability. For example, integrating road and weather data can enhance route planning and safety in adverse conditions[77]. Another task is predicting material properties and performance to support vehicle design. By fusing structural, experimental and supply-chain data, multimodal AI can optimize materials for lightweight, durable and cost-effective vehicles, aligning with sustainability goals[78]. Additional tasks include passenger-focused objectives, such as integrating biosignals and cabin sensors to enhance comfort and safety. Ensuring privacy compliance remains integral across all tasks.

Developing multimodal AI systems for self-driving cars follows the five-step workflow above, adapted to the demands of autonomous systems. Data collection involves assembling sensor inputs (for example, LiDAR, radar, cameras) and materials data to build comprehensive datasets. Data curation addresses challenges such as synchronizing asynchronous sensor inputs and standardizing materials information. Learning can use fusion strategies to enhance system reliability, while physics-informed neural networks[79,80] can integrate scientific laws to improve realistic simulations of driving scenarios and materials development. Evaluation ensures robustness through multimetric validation across diverse scenarios, including urban environments and extreme weather conditions. Finally, interpretation in autonomous systems supports real-time explainability and transparency, enabling stakeholders to trust and audit system decisions. By aligning these steps with the unique requirements of autonomous systems, multimodal AI can ensure adaptability, safety and performance.

Real-world deployment of multimodal AI systems for self-driving cars requires live sensor integration, infrastructure readiness and operational resilience across diverse environments. Robust in-vehicle computing and inter-vehicle communication enable reliable and coordinated performance at scale. Assessment validates multimodal AI's added value through simulations and multicentre trials. Testing performance under extreme conditions, such as adverse weather or high-traffic intersections, ensures reliability. Addressing scalability and privacy compliance further supports ethical, sustainable deployment, advancing societal trust in self-driving technologies[81].

Climate change adaptation

Climate change adaptation (Fig. 4c) involves forecasting extreme weather, assessing climate risks and managing resources to minimize environmental and socio-economic impacts[82–85]. Defining the problem involves assessing whether multimodal AI can surpass unimodal approaches by integrating diverse data sources for deeper insights and actionable strategies. For example, multimodal AI can combine satellite imagery, historical climate records and geospatial data to model weather patterns and predict extreme events, supporting proactive disaster management[86]. Integrating socio-economic data with environmental observations can enable more comprehensive climate risk assessments[87].

Deploying multimodal AI for climate change adaptation must navigate privacy, technical and economic constraints. Privacy concerns arise from integrating sensitive data, such as socio-economic records and proprietary satellite imagery. Ensuring compliance with global and regional privacy regulations and the ethical use of data supports responsible deployment. Technical challenges include the heterogeneity and sparsity of environmental and socio-economic data. Economic constraints, such as the high cost of advanced sensors, necessitate balancing low-cost and low-fidelity data sources with high-cost, high-fidelity alternatives to maintain accessibility and scalability. By addressing these constraints, multimodal AI can deliver equitable and sustainable climate solutions in diverse global contexts.

For climate change adaptation, task formulation aligns complex environmental challenges with actionable AI tasks. A primary task is forecasting extreme weather events by integrating multimodal data such as satellite imagery, historical climate records and real-time observations. These forecasts can enable timely interventions to

 https://doi.org/10.1038/s42256-025-01116-5

BOX 1

Strategic recommendations for advancing multimodal AI across disciplines

Deployment-centric development

- Safety, reliability, and interpretability
  Challenge: Ensuring reliable, safe deployment across varied conditions and building user trust to facilitate understanding and broader adoption.
  Recommendation: Translate complex data and outputs into accessible formats (images, text or speech); incorporate domain-specific knowledge; implement human-in-the-loop systems for verification and validation; conduct rigorous usability testing; develop standards for safety, reliability and interpretability criteria; enforce such criteria in research and peer review[36].
- Scalability and resource efficiency
  Challenge: Scaling multimodal AI to handle vast data volumes while optimizing resource consumption, maintaining system availability and ensuring uninterrupted communication between system components.
  Recommendation: Innovate scalable and resource-efficient multimodal AI solutions; promote collaboration between AI developers and hardware providers for sustainable resource use; ensure stable system availability and seamless intercomponent communication; develop cloud-based solutions and edge computing to power scalability.
- Ethical compliance and user preparedness
  Challenge: Addressing privacy, consent and bias in applications with profound societal impacts.
  Recommendation: Develop clear ethical guidelines; include end-users in the design; involve human oversight and create user feedback loops for continuous solution refinement; provide comprehensive training for reliable deployment.

Data-centric development

- Data scarcity and access
  Challenge: Limited availability of high-quality multimodal datasets spanning a comprehensive range of modalities, complicated further by privacy and ethical constraints and additional heterogeneity when data sharing spans secure data environments, organizations and regions.
  Recommendation: Promote open data initiatives, such as anonymized data-sharing programmes; invest in cross-discipline efforts for data collection and curation to improve availability; build data infrastructures that ensure secure, privacy-compliant and ethics-compliant access across domains and regions; establish guidelines to standardize formats and annotations; develop advanced tools such as knowledge graphs to integrate structured and unstructured data sources for improved contextual understanding, reasoning and interoperability.
- Balanced data representation
  Challenge: Bias due to the limited availability of data representing underrepresented or less-studied categories (for example, populations and regions).
  Recommendation: Diversify dataset development to enhance model generalizability and reduce bias; build and enhance multimodal data platforms (for example, UK Biobank[128], MIMIC[129], Materials Project[120], ERA5[123]) to improve global data accessibility and quality.

Model-centric development

- Multimodal fusion and modality selection
  Challenge: Managing diverse properties and patterns across modalities, including avoiding unnecessary modality redundancy, to maintain integrity and utility while enhancing integration.
  Recommendation: Assess the value of additional modalities by comparing unimodal, few-modality and full-modality models; enhance semantic modality alignment—for example, via large language models; develop guidelines for optimal modality selection aligned with deployment context and cost–performance constraints; address other multimodal-AI-specific modelling challenges (Fig. 3b).
- Foundation models
  Challenge: Developing foundation models beyond vision and language requires substantial resources, limiting accessibility across disciplines.
  Recommendation: Develop and expand multimodal foundation models beyond vision and language as critical reusable infrastructure to lower barriers; invest in downstream research and advance methods to reduce resource costs—for example, using synthetic or weakly labelled data to ease training demands[44]; pair underrepresented modalities (for example, graphs) with more interpretable ones (for example, language) to improve usability and broaden accessibility for non-expert stakeholders.

Stakeholder engagement and interdisciplinary collaboration

- Stakeholder inclusion and alignment
  Challenge: Limited involvement of stakeholders (for example, domain experts, end-users and regulators) during early-stage planning can result in solutions that are technically sound but poorly aligned with deployment contexts.
  Recommendation: Integrate stakeholder input across the AI lifecycle through co-design, participatory planning and regulatory consultation to ensure contextual relevance, trust and successful real-world adoption.
- Cross-disciplinary standards and communication
  Challenge: Discrepancies in multimodal data standards and communication barriers across disciplines.
  Recommendation: Promote cross-disciplinary standards for multimodal data integration; organize interdisciplinary exchange events to align goals, foster collaboration and build a thriving ecosystem.
- Intellectual property and workflow adaptation
  Challenge: Intellectual property concerns and resistance to change established workflows.
  Recommendation: Develop adaptive strategies and open collaborative platforms to facilitate cross-disciplinary projects.

mitigate human and economic losses[88], including early warning systems for floods that integrate geospatial, hydrological and social data streams[89]. Another task is predicting the socio-economic impacts of severe weather by combining spatial, market and socio-economic data, guiding resource allocation and policy decisions[90]. These tasks align multimodal AI with both technical requirements and societal priorities, advancing climate resilience and equitable adaptation strategies.

Developing multimodal AI systems for climate change adaptation follows a similar five-step workflow. Data collection gathers remote sensing data, near-real-time sensor network outputs, historical climate records and socio-economic metrics, creating comprehensive datasets for tasks such as extreme weather forecasting and early warning systems. Data curation addresses challenges such as temporal misalignment and data sparsity by standardizing inputs and enriching sparse data through interpolation or integration of auxiliary sources. Multimodal learning can benefit from advanced models such as Aurora[91], which fuse satellite imagery and meteorological data to improve atmospheric predictions. Evaluation uses multimetric benchmarks to test system performance in diverse scenarios. For example, WeatherBench 2 provides a standard platform for assessing atmospheric prediction models[92]. Interpretation ensures that outputs are transparent and actionable for stakeholders, supporting evidence-based decision-making. By aligning these steps with climate-specific challenges, multimodal AI can deliver reliable, scalable and adaptable solutions.

Real-world deployment of multimodal AI for climate change adaptation translates research insights into scalable, operational systems. Early-warning platforms and impact forecasting models must synchronize diverse data streams, deliver real-time outputs and integrate with decision-making infrastructures across sectors and regions[93,94]. Real-time monitoring requires robust infrastructure to synchronize and process diverse modalities. Deployment challenges include latency, data coverage gaps and variable infrastructure capacity across geographies. Assessment involves stress testing and multicentre validation to ensure reliability across environmental and socio-economic scenarios. Ethical considerations, such as prioritizing vulnerable communities and ensuring equitable access to data and AI tools, help enable sustainable deployment. By overcoming deployment challenges, multimodal AI systems can empower policy-makers, industries and communities to respond proactively to climate challenges, advancing global sustainability efforts.

Outlook

Most existing research on multimodal AI has focused on model-centric development. While attention to data-centric development is increasing, unlocking the full potential of multimodal AI and addressing real-world challenges will ultimately require a shift towards deployment-centric development. This shift will require strategic advancements in data, model and deployment methods, as well as concerted efforts to address multimodal-AI-specific challenges, namely modality incompleteness, multimodal heterogeneity, cross-modality alignment, modality complementarity and multimodal privacy risk, through stakeholder engagement, interdisciplinary collaborations and community-building. Box 1 presents strategic recommendations to advance multimodal AI across disciplines under a deployment-centric perspective. These recommendations draw on insights from the three use cases and the broader cross-disciplinary analysis in Supplementary Information on multimodal AI beyond these three use cases, providing actionable guidance for future developments.

Deployment-centric development brings challenges around safety, reliability, interpretability, scalability and ethics, particularly as multimodal AI expands in high-stakes applications such as healthcare and sustainability. Addressing these challenges requires implementing robust human-in-the-loop systems, developing clear standards for safety and transparency, and prioritizing scalable, resource-efficient infrastructure to support increasing demands for data integration, processing and storage.

Robust data-centric development underpins deployment-centric progress by ensuring data availability, diversity and quality. High-quality multimodal data are often limited, and existing datasets often lack the diversity needed to ensure fairness in AI systems. Clear benchmarks[95,96] and globally accessible datasets (Fig. 5) are needed to enhance trust, consistency and adaptability across disciplines, facilitating reproducible research. Promoting secure data-sharing frameworks[97], open initiatives such as the European Life Science Infrastructure for Biological Information[98] and globally accessible multimodal data platforms will further strengthen AI's capacity to address diverse global challenges. Knowledge graphs[99] also offer promise for organizing varied data formats to unify and contextualize multimodal inputs across disciplines.

Model-centric development for multimodal AI faces unique challenges in effectively and efficiently fusing diverse modalities. While foundation models have proved effective for vision and language tasks, expanding them to other modalities and disciplines[100] could substantially lower development barriers. Guidelines and frameworks for selecting relevant modalities, comparing multimodal versus unimodal performance, and evaluating the benefits of using many versus few modalities will help maximize model efficiency and utility. As large language models, multimodal foundation models and generative AI systems grow in prominence[101–104], deployment-centric design is having an increasingly important role in guiding architectural choices, managing inference-time costs and aligning with data and domain-specific constraints[105,106]. These models also increasingly rely on synthetic or weakly labelled data, or cross-modal supervision to reduce annotation costs[64,107–111], reinforcing the need for deployment-aware data strategies.

Stakeholder engagement and interdisciplinary collaboration are both crucial for progress, as illustrated across the three use cases. Integrating stakeholder input ensures contextual relevance and trust. Standardizing data practices, fostering cross-disciplinary knowledge exchange and developing shared platforms help bridge disciplinary gaps and enable multimodal AI to address complex societal challenges more effectively and holistically[36,37].

Building a dynamic and inclusive multimodal AI community fosters innovation and drives solutions to complex real-world challenges. Collaborative efforts, through regular workshops, forums and interdisciplinary research initiatives, facilitate knowledge exchange and inspire collective problem-solving across disciplines. Engaging researchers, practitioners and stakeholders from diverse backgrounds, particularly early-career researchers and underrepresented groups, will not only enrich the AI ecosystem but also broaden the perspectives and expertise shaping it, making it more 'multimodal'. Community-driven initiatives[112,113], including comprehensive surveys or perspective papers, provide insights that guide future developments, ensuring multimodal AI advances in responsible, impactful directions.

Ultimately, the success of multimodal AI lies in its deployment. By embracing a deployment-centric mindset, the community can turn research breakthroughs into real-world impact across disciplines.

Data availability

Source data are provided with this paper. They are also available at https://github.com/multimodalAI/multimodal-ai-landscape, where they will be updated annually.

References

1. Wang, H. et al. Scientific discovery in the age of artificial intelligence. *Nature* **620**, 47–60 (2023).
2. Kabir, M. M., Jim, J. R. & Istenes, Z. Terrain detection and segmentation for autonomous vehicle navigation: a state-of-the-art systematic review. *Inf. Fusion* **113**, 102644 (2025).

3. Woloshin, S., Patel, N. & Kesselheim, A. S. False negative tests for SARS-CoV-2 infection—challenges and implications. *N. Engl. J. Med.* **383**, e38 (2020).
4. Zhao, F., Zhang, C. & Geng, B. Deep multimodal data fusion. *ACM Comput. Surv.* **56**, 216 (2024).
5. Baltrušaitis, T., Ahuja, C. & Morency, L.-P. Multimodal machine learning: a survey and taxonomy. *IEEE Trans. Pattern Anal. Mach. Intell.* **41**, 423–443 (2019).
6. Ektefaie, Y., Dasoulas, G., Noori, A., Farhat, M. & Zitnik, M. Multimodal learning with graphs. *Nat. Mach. Intell.* **5**, 340–350 (2023).
7. Xu, P., Zhu, X. & Clifton, D. A. Multimodal learning with transformers: a survey. *IEEE Trans. Pattern Anal. Mach. Intell.* **45**, 12113–12132 (2023).
8. Liang, P. P., Zadeh, A. & Morency, L.-P. Foundations & trends in multimodal machine learning: principles, challenges, and open questions. *ACM Comput. Surv.* **56**, 264 (2024).
9. Acosta, J. N., Falcone, G. J., Rajpurkar, P. & Topol, E. J. Multimodal biomedical AI. *Nat. Med.* **28**, 1773–1784 (2022).
10. Kline, A. et al. Multimodal machine learning in precision health: a scoping review. *npj Digit. Med.* **5**, 171 (2022).
11. Krones, F., Marikkar, U., Parsons, G., Szmul, A. & Mahdi, A. Review of multimodal machine learning approaches in healthcare. *Inf. Fusion* **114**, 102690 (2025).
12. Notin, P., Rollins, N., Gal, Y., Sander, C. & Marks, D. Machine learning for functional protein design. *Nat. Biotechnol.* **42**, 216–228 (2024).
13. Song, B., Zhou, R. & Ahmed, F. Multi-modal machine learning in engineering design: a review and future directions. *J. Comput. Inf. Sci. Eng.* **24**, 010801 (2024).
14. Ofodile, O. C. et al. Predictive analytics in climate finance: assessing risks and opportunities for investors. *GSC Adv. Res. Rev.* **18**, 423–433 (2024).
15. Quatrini, S. Challenges and opportunities to scale up sustainable finance after the COVID-19 crisis: lessons and promising innovations from science and practice. *Ecosyst. Serv.* **48**, 101240 (2021).
16. Gupta, V. et al. An emotion care model using multimodal textual analysis on COVID-19. *Chaos Solitons Fractals* **144**, 110708 (2021).
17. Anshul, A., Pranav, G. S., Zia Ur Rehman, M. & Kumar, N. A multimodal framework for depression detection during COVID-19 via harvesting social media. *IEEE Trans. Comput. Soc. Syst.* **11**, 2872–2888 (2024).
18. Bordes, F. et al. An introduction to vision–language modeling. Preprint at https://doi.org/10.48550/arXiv.2405.17247 (2024).
19. Zhang, J., Huang, J., Jin, S. & Lu, S. Vision–language models for vision tasks: a survey. *IEEE Trans. Pattern Anal. Mach. Intell.* **46**, 5625–5644 (2024).
20. van Breugel, B. & van der Schaar, M. Position: why tabular foundation models should be a research priority. In *Proc. 41st International Conference on Machine Learning* (eds Salakhutdinov, R. et al.) 48976–48993 (PMLR, 2024).
21. Ma, M. et al. SMIL: multimodal learning with severely missing modality. In *Proc. AAAI Conference on Artificial Intelligence* Vol. 35, 2302–2310 (AAAI Press, 2021).
22. Wu, R., Wang, H., Chen, H.-T. & Carneiro, G. Deep multimodal learning with missing modality: a survey. Preprint at https://doi.org/10.48550/arXiv.2409.07825 (2024).
23. Wang, F., Zhou, Y., Wang, S., Vardhanabhuti, V. & Yu, L. Multi-granularity cross-modal alignment for generalized medical visual representation learning. *Adv. Neural Inf. Process. Syst.* **35**, 33536–33549 (2022).
24. Zhao, T., Zhang, L., Ma, Y. & Cheng, L. A survey on safe multi-modal learning systems. In *Proc. 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining* 6655–6665 (Association for Computing Machinery, 2024).
25. Pranjal, R. et al. Toward privacy-enhancing ambulatory-based well-being monitoring: investigating user re-identification risk in multimodal data. In *IEEE International Conference on Acoustics, Speech and Signal Processing* 1–5 (IEEE, 2023).
26. Paleyes, A., Urma, R.-G. & Lawrence, N. D. Challenges in deploying machine learning: a survey of case studies. *ACM Comput. Surv.* **55**, 114 (2022).
27. Jo, A. The promise and peril of generative AI. *Nature* **614**, 214–216 (2023).
28. Brown, T. et al. Language models are few-shot learners. In *Proc. 34th International Conference on Neural Information Processing Systems* (eds Larochelle, H. et al.) 1877–1901 (Curran Associates Inc., 2020).
29. Achiam, J. et al. GPT-4 technical report. Preprint at https://doi.org/10.48550/arXiv.2303.08774 (2023).
30. Seedat, N., Imrie, F. & van der Schaar, M. Navigating data-centric artificial intelligence with DC-Check: advances, challenges, and opportunities. *IEEE Trans. Artif. Intell.* **5**, 2589–2603 (2024).
31. Liu, H., Li, C., Wu, Q. & Lee, Y. J. Visual instruction tuning. In *Proc. 37th International Conference on Neural Information Processing Systems* (eds Oh, A. et al.) 34892–34916 (Curran Associates Inc., 2023).
32. Li, C. et al. LLaVA-med: training a large language-and-vision assistant for biomedicine in one day. In *Proc. 37th International Conference on Neural Information Processing Systems* (eds Oh, A. et al.) 28541–28564 (Curran Associates Inc., 2023).
33. Roberts, M. et al. Common pitfalls and recommendations for using machine learning to detect and prognosticate for COVID-19 using chest radiographs and CT scans. *Nat. Mach. Intell.* **3**, 199–217 (2021).
34. Kreuzberger, D., Kühl, N. & Hirschl, S. Machine learning operations (MLOps): overview, definition, and architecture. *IEEE Access* **11**, 31866–31879 (2023).
35. Lavin, A. et al. Technology readiness levels for machine learning systems. *Nat. Commun.* **13**, 6039 (2022).
36. Nielsen, M. W. et al. Intersectional analysis for science and technology. *Nature* **640**, 329–337 (2025).
37. Lekadir, K. et al. FUTURE-AI: international consensus guideline for trustworthy and deployable artificial intelligence in healthcare. *BMJ* **388**, e081554 (2025).
38. Huang, Y. et al. What makes multi-modal learning better than single (provably). In *Proc. 35th International Conference on Neural Information Processing Systems* (eds Ranzato, M. et al.) 10944–10956 (Curran Associates Inc., 2021).
39. Meng, X., Babaee, H. & Karniadakis, G. E. Multi-fidelity Bayesian neural networks: algorithms and applications. *J. Comput. Phys.* **438**, 110361 (2021).
40. Penwarden, M., Zhe, S., Narayan, A. & Kirby, R. M. Multifidelity modeling for physics-informed neural networks (PINNs). *J. Comput. Phys.* **451**, 110844 (2022).
41. Zitnik, M. et al. Machine learning for integrating data in biology and medicine: principles, practice, and opportunities. *Inf. Fusion* **50**, 71–91 (2019).
42. Lunke, S. et al. Integrated multi-omics for rapid rare disease diagnosis on a national scale. *Nat. Med.* **29**, 1681–1691 (2023).
43. Wu, E. et al. How medical AI devices are evaluated: limitations and recommendations from an analysis of FDA approvals. *Nat. Med.* **27**, 582–584 (2021).
44. van Breugel, B., Liu, T., Oglic, D. & van der Schaar, M. Synthetic data in biomedicine via generative artificial intelligence. *Nat. Rev. Bioeng.* **2**, 991–1004 (2024).
45. Al-Rubaie, M. & Chang, J. M. Privacy-preserving machine learning: threats and solutions. *IEEE Secur. Priv.* **17**, 49–58 (2019).
46. Wendland, P. et al. Generation of realistic synthetic data using multimodal neural ordinary differential equations. *npj Digit. Med.* **5**, 122 (2022).

47. Che, L., Wang, J., Zhou, Y. & Ma, F. Multimodal federated learning: a survey. *Sensors* **23**, 6986 (2023).
48. Wang, Q., Zhan, L., Thompson, P. M. & Zhou, J. Multimodal learning with incomplete modalities by knowledge distillation. In *Proc. 26th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining* 1828–1838 (Association for Computing Machinery, 2020).
49. Yu, Z., Yu, J., Fan, J. & Tao, D. Multi-modal factorized bilinear pooling with co-attention learning for visual question answering. In *Proc. IEEE International Conference on Computer Vision* 1839–1848 (IEEE, 2017).
50. Weidinger, L. et al. Holistic safety and responsibility evaluations of advanced AI models. Preprint at https://doi.org/10.48550/arXiv.2404.14068 (2024).
51. Luccioni, S., Akiki, C., Mitchell, M. & Jernite, Y. Stable bias: evaluating societal representations in diffusion models. In *Proc. 37th International Conference on Neural Information Processing Systems* (eds Oh, A. et al.) 56338–56351 (Curran Associates Inc., 2023).
52. Murdoch, W. J., Singh, C., Kumbier, K., Abbasi-Asl, R. & Yu, B. Definitions, methods, and applications in interpretable machine learning. *Proc. Natl Acad. Sci. USA* **116**, 22071–22080 (2019).
53. Rudin, C. Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. *Nat. Mach. Intell.* **1**, 206–215 (2019).
54. Gawlikowski, J. et al. A survey of uncertainty in deep neural networks. *Artif. Intell. Rev.* **56**, 1513–1589 (2023).
55. Khojasteh, D., Davani, E., Shamsipour, A., Haghani, M. & Glamore, W. Climate change and COVID-19: interdisciplinary perspectives from two global crises. *Sci. Total Environ.* **844**, 157142 (2022).
56. Elmer, T., Mepham, K. & Stadtfeld, C. Students under lockdown: comparisons of students' social networks and mental health before and during the COVID-19 crisis in Switzerland. *PLoS ONE* **15**, e0236337 (2020).
57. Valdez, D., Ten Thij, M., Bathina, K., Rutter, L. A. & Bollen, J. Social media insights into US mental health during the COVID-19 pandemic: longitudinal analysis of Twitter data. *J. Med. Internet Res.* **22**, e21418 (2020).
58. Liu, Z. et al. Near-real-time monitoring of global CO_2 emissions reveals the effects of the COVID-19 pandemic. *Nat. Commun.* **11**, 5172 (2020).
59. Zheng, B. et al. Satellite-based estimates of decline and rebound in China's CO_2 emissions during COVID-19 pandemic. *Sci. Adv.* **6**, eabd4998 (2020).
60. Mosser, P. C. Central bank responses to COVID-19. *Bus. Econ.* **55**, 191–201 (2020).
61. Nicola, M. et al. The socio-economic implications of the coronavirus pandemic (COVID-19): a review. *Int. J. Surg.* **78**, 185–193 (2020).
62. Ding, X. et al. Wearable sensing and telehealth technology with potential applications in the coronavirus pandemic. *IEEE Rev. Biomed. Eng.* **14**, 48–70 (2020).
63. Charlton, P. H. et al. Wearable photoplethysmography for cardiovascular monitoring. *Proc. IEEE* **110**, 355–381 (2022).
64. Zong, Y., Mac Aodha, O. & Hospedales, T. Self-supervised multimodal learning: a survey. *IEEE Trans. Pattern Anal. Mach. Intell.* **47**, 5299-5318 (2025).
65. Yang, X., Zhang, T. & Xu, C. Cross-domain feature learning in multimedia. *IEEE Trans. Multimed.* **17**, 64–78 (2014).
66. Han, R. et al. Randomised controlled trials evaluating artificial intelligence in clinical practice: a scoping review. *Lancet Digit. Health* **6**, e367–e373 (2024).
67. Plana, D. et al. Randomized clinical trials of machine learning interventions in health care: a systematic review. *JAMA Netw. Open* **5**, e2233946 (2022).
68. Imrie, F., Davis, R. & van der Schaar, M. Multiple stakeholders drive diverse interpretability requirements for machine learning in healthcare. *Nat. Mach. Intell.* **5**, 824–829 (2023).
69. Mincu, D. & Roy, S. Developing robust benchmarks for driving forward AI innovation in healthcare. *Nat. Mach. Intell.* **4**, 916–921 (2022).
70. Soni, A. et al. in *Machine Learning for Robotics Applications* (eds Bianchini, M. et al.) 139–151 (Springer, 2021).
71. Badue, C. et al. Self-driving cars: a survey. *Expert Syst. Appl.* **165**, 113816 (2021).
72. Yeong, D. J., Velasco-Hernandez, G., Barry, J. & Walsh, J. Sensor and sensor fusion technology in autonomous vehicles: a review. *Sensors* **21**, 2140 (2021).
73. Yang, J. et al. Generalized predictive model for autonomous driving. In *Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition* 14662–14672 (IEEE, 2024).
74. Zhang, W. & Xu, J. Advanced lightweight materials for automobiles: a review. *Mater. Des.* **221**, 110994 (2022).
75. Hansson, S. O., Belin, M.-Å. & Lundgren, B. Self-driving vehicles—an ethical overview. *Philos. Technol.* **34**, 1383–1408 (2021).
76. Chowdhury, A., Karmakar, G., Kamruzzaman, J., Jolfaei, A. & Das, R. Attacks on self-driving cars and their countermeasures: a survey. *IEEE Access* **8**, 207308–207342 (2020).
77. Dey, K. C., Mishra, A. & Chowdhury, M. Potential of intelligent transportation systems in mitigating adverse weather impacts on road mobility: a review. *IEEE Trans. Intell. Transp. Syst.* **16**, 1107–1119 (2014).
78. Kamran, S. S. et al. Artificial intelligence and advanced materials in automotive industry: potential applications and perspectives. *Mater. Today Proc.* **62**, 4207–4214 (2022).
79. Raissi, M., Perdikaris, P. & Karniadakis, G. E. Physics-informed neural networks: a deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *J. Comput. Phys.* **378**, 686–707 (2019).
80. Karniadakis, G. E. et al. Physics-informed machine learning. *Nat. Rev. Phys.* **3**, 422–440 (2021).
81. Liu, P., Yang, R. & Xu, Z. Public acceptance of fully automated driving: effects of social trust and risk/benefit perceptions. *Risk Anal.* **39**, 326–341 (2019).
82. Bi, K. et al. Accurate medium-range global weather forecasting with 3D neural networks. *Nature* **619**, 533–538 (2023).
83. Lam, R. et al. Learning skillful medium-range global weather forecasting. *Science* **382**, 1416–1421 (2023).
84. Mathiesen, K. Rating climate risks to credit worthiness. *Nat. Clim. Change* **8**, 454–456 (2018).
85. Breitenstein, M., Ciummo, S. & Walch, F. *Disclosure of Climate Change Risk in Credit Ratings*. Occasional Paper Series 303 (European Central Bank, 2022).
86. Imran, M., Ofli, F., Caragea, D. & Torralba, A. Using AI and social media multimodal content for disaster response and management: opportunities, challenges, and future directions. *Inf. Process. Manag.* **57**, 102261 (2020).
87. Tanir, T., Yildirim, E., Ferreira, C. M. & Demir, I. Social vulnerability and climate risk assessment for agricultural communities in the United States. *Sci. Total Environ.* **908**, 168346 (2024).
88. Kumar, P. et al. An overview of monitoring methods for assessing the performance of nature-based solutions against natural hazards. *Earth Sci. Rev.* **217**, 103603 (2021).
89. Tkachenko, N., Jarvis, S. & Procter, R. Predicting floods with Flickr tags. *PLoS ONE* **12**, e0172870 (2017).
90. Thulke, D. et al. ClimateGPT: towards AI synthesizing interdisciplinary research on climate change. Preprint at https://doi.org/10.48550/arXiv.2401.09646 (2024).
91. Bodnar, C. et al. *Aurora: a Foundation Model of the Atmosphere*. Technical Report MSR-TR-2024-16 (Microsoft Research AI for Science, 2024).

92. Rasp, S. et al. WeatherBench 2: a benchmark for the next generation of data-driven global weather models. *J. Adv. Model. Earth Syst.* **16**, e2023MS004019 (2024).
93. Morshed, S. R. et al. Decoding seasonal variability of air pollutants with climate factors: a geostatistical approach using multimodal regression models for informed climate change mitigation. *Environ. Pollut.* **345**, 123463 (2024).
94. Reichstein, M. et al. Early warning of complex climate risk with integrated artificial intelligence. *Nat. Commun.* **16**, 2564 (2025).
95. Liang, P. P. et al. MultiBench: multiscale benchmarks for multimodal representation learning. In *Proc. 35th Conference on Neural Information Processing Systems Track on Datasets and Benchmarks* (eds Vanschoren, J. & Yeung, S.) (2021)
96. de la Fuente, J. et al. Towards a more inductive world for drug repurposing approaches. *Nat. Mach. Intell.* **7**, 495–508 (2025).
97. Torabi, F. et al. The common governance model: a way to avoid data segregation between existing trusted research environment. *Int. J. Popul. Data Sci.* **8**, 2164 (2023).
98. Crosswell, L. C. & Thornton, J. M. ELIXIR: a distributed infrastructure for European biological data. *Trends Biotechnol.* **30**, 241–242 (2012).
99. Liang, K. et al. A survey of knowledge graph reasoning on graph types: static, dynamic, and multi-modal. *IEEE Trans. Pattern Anal. Mach. Intell.* **46**, 9456–9478 (2024).
100. Archit, A. et al. Segment anything for microscopy. *Nat. Methods* **22**, 579–591 (2025).
101. Li, C. et al. Multimodal foundation models: from specialists to general-purpose assistants. *Found. Trends Comput. Graph. Vis.* **16**, 1–214 (2024).
102. Fei, N. et al. Towards artificial general intelligence via a multimodal foundation model. *Nat. Commun.* **13**, 3094 (2022).
103. Narayanswamy, G. et al. Scaling wearable foundation models. In *The Thirteenth International Conference on Learning Representations* (ICLR, 2025).
104. Cui, H. et al. Towards multimodal foundation models in molecular cell biology. *Nature* **640**, 623–633 (2025).
105. Introducing the Model Context Protocol. *Anthropic* https://www.anthropic.com/index/model-context-protocol (2024).
106. Weisz, J. D. et al. Design principles for generative AI applications. In *Proc. 2024 CHI Conference on Human Factors in Computing Systems* (eds Mueller, F. F. et al.) 378 (Association for Computing Machinery, 2024).
107. Li, J., Li, D., Xiong, C. & Hoi, S. BLIP: Bootstrapping language–image pre-training for unified vision–language understanding and generation. In *Proc. 39th International Conference on Machine Learning* (eds Chaudhuri, K. et al.) 12888–12900 (PMLR, 2022).
108. Li, J., Li, D., Savarese, S. & Hoi, S. BLIP-2: Bootstrapping language–image pre-training with frozen image encoders and large language models. In *Proc. 40th International Conference on Machine Learning* (eds Krause, A. et al.) 19730–19742 (PMLR, 2023).
109. Driess, D. et al. PaLM-E: an embodied multimodal language model. In *Proc. 40th International Conference on Machine Learning* (eds Krause, A. et al.) 8469–8488 (PMLR, 2023).
110. Tan, Z. et al. Large language models for data annotation and synthesis: a survey. In *Proc. 2024 Conference on Empirical Methods in Natural Language Processing* (eds Al-Onaizan, Y. et al.) 930–957 (Association for Computational Linguistics, 2024).
111. Ding, B. et al. Data augmentation using LLMs: data perspectives, learning paradigms and challenges. In *Findings of the Association for Computational Linguistics ACL 2024* (eds Ku, L.-W. et al.) 1679–1705 (Association for Computational Linguistics, 2024).
112. Chan, A.-W. et al. Reporting guidelines for clinical trial reports for interventions involving artificial intelligence: the CONSORT-AI extension. *Nat. Med.* **26**, 1364–1374 (2020).
113. de Hond, A. A. H. et al. Guidelines and quality criteria for artificial intelligence-based prediction models in healthcare: a scoping review. *npj Digit. Med.* **5**, 2 (2022).
114. Tomczak, K., Czerwińska, P. & Wiznerowicz, M. The Cancer Genome Atlas (TCGA): an immeasurable source of knowledge. *Contemp. Oncol.* **2015**, 68–77 (2015).
115. Johnson, A. E. W. et al. MIMIC-IV, a freely accessible electronic health record dataset. *Sci. Data* **10**, 1 (2023).
116. Tong, E., Zadeh, A., Jones, C. & Morency, L.-P. Combating human trafficking with multimodal deep models. In *Proc. 55th Annual Meeting of the Association for Computational Linguistics* (eds Barzilay, R. et al.) 1547–1556 (Association for Computational Linguistics, 2017).
117. Razi, A. et al. Instagram data donation: a case study on collecting ecologically valid social media data for the purpose of adolescent online risk detection. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts* (eds Barbosa, S. et al.) 1–9 (Association for Computing Machinery, 2022).
118. Caesar, H. et al. nuScenes: a multimodal dataset for autonomous driving. In *Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition* 11621–11631 (IEEE, 2020).
119. Yu, H. et al. DAIR-V2X: a large-scale dataset for vehicle–infrastructure cooperative 3D object detection. In *Proc. IEEE/CVF Conference on Computer Vision and Pattern Recognition* 21361–21370 (IEEE, 2022).
120. Jain, A. et al. Commentary: The Materials Project: a materials genome approach to accelerating materials innovation. *APL Mater.* **1**, 011002 (2013).
121. Lu, P. et al. MathVista: Evaluating mathematical reasoning of foundation models in visual contexts. In *International Conference on Learning Representations* (ICLR, 2024).
122. iNaturalist contributors. iNaturalist research-grade observations. *GBIF* https://doi.org/10.15468/ab3s5x (2025).
123. Hersbach, H. et al. The ERA5 global reanalysis. *Q. J. R. Meteorol. Soc.* **146**, 1999–2049 (2020).
124. Mathur, P. et al. Monopoly: financial prediction from monetary policy conference videos using multimodal cues. In *Proc. 30th ACM International Conference on Multimedia* 2276–2285 (Association for Computing Machinery, 2022).
125. Cheng, D., Yang, F., Xiang, S. & Liu, J. Financial time series forecasting with multi-modality graph neural network. *Pattern Recognit.* **121**, 108218 (2022).
126. Chen, K. et al. The operational medium-range deterministic weather forecasting can be extended beyond a 10-day lead time. *Commun. Earth Environ.* **6**, 518 (2025).
127. Jin, Q. et al. Spatiotemporal inference network for precipitation nowcasting with multimodal fusion. *IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens.* **17**, 1299–1314 (2024).
128. Bycroft, C. et al. The UK Biobank resource with deep phenotyping and genomic data. *Nature* **562**, 203–209 (2018).
129. Johnson, A. E. W. et al. MIMIC-III, a freely accessible critical care database. *Sci. Data* **3**, 160035 (2016).

Acknowledgements

This work was enabled and supported by the Alan Turing Institute. We thank T. Chakraborty and C. Li for inspiring this work, D. A. Clifton for his support and T. Dunstan for contributing to the climate change adaptation section. J.Z. is supported by donations from D. Naik and S. Naik. S.Z. is supported by EPSRC (grant EP/Y017544/1). T.L.v.d.P. was supported by EPSRC (grant EP/Y028880/1). A.V. is supported by UKRI CDT in AI for Healthcare (grant EP/S023283/1). A.G. is supported by the Research Council of Norway (secureIT project 288787). M.Z. is supported by EPSRC (grant EP/X031276/1). C.T. is supported by UKRI CDT in AI-enabled Healthcare (grant EP/S021612/1). R.L. is supported by the Royal Society (grant IEC\NSFC\233558). L.v.Z. is supported by NERC (grant NE/W004747/1). O.T. is supported by UKRI CDT in Application of Artificial Intelligence to the study of Environmental Risks (grant EP/S022961/1). Z.A.S. is supported by Google DeepMind. O.A. is supported by NIHR Barts BRC (grant NIHR203330). T.N.D. is supported by UKRI CDT in Accountable, Responsible and Transparent AI (grant EP/S023437/1). L.F. is supported by MRC (grant MR/W006804/1). N.J. is supported by the EU's co-funded HE project MuseIT (grant 101061441). M.V. is supported by St George's Hospital Charity. A.C.-C. is supported by EPSRC (grant EP/Y028880/1). H.W. is supported by MRC (grant MR/X030075/1). C.C. is supported by the Royal Society (grant GS\R2\242355). T.Z. was supported by the Royal Academy of Engineering (grant RF\201819\18\109). G.G.S. is supported by EPSRC (grant EP/Y009800/1). P.H.C. is supported by BHF (grant FS/20/20/34626). H.L. is supported by EPSRC (grant UKRI396). The views expressed in this material are those of the authors and do not necessarily represent the views of their affiliated institutions or funders.

Author contributions

X.L., J.Z., S.Z., T.L.v.d.P., S.T., T.Z., W.K.C., P.H.C. and H.L. conceptualized the manuscript. X.L., J.Z., S.Z., A.G., M.Z., L.v.Z., O.T. and H.L. designed the figures. X.L. and H.L. coordinated the entire project. All authors contributed to writing, resources or editing.

Competing interests

G.G.S. is a scientific advisory board member at BioAI Health. P.H.C. provides consulting services to Cambridge University Technical Services for wearable manufacturers. The remaining authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s42256-025-01116-5.

Correspondence should be addressed to Haiping Lu.

Peer review information *Nature Machine Intelligence* thanks Girmaw Abebe Tadesse and the other, anonymous, reviewer(s) for their contribution to the peer review of this work.

Xianyuan Liu [1,32], Jiayang Zhang [1,32], Shuo Zhou [2,32], Thijs L. van der Plas [3], Avish Vijayaraghavan [4], Anastasiia Grishina [5], Mengdie Zhuang [6], Daniel Schofield [7], Christopher Tomlinson [8], Yuhan Wang [9], Ruizhe Li [10], Louisa van Zeeland [3], Sina Tabakhi [2], Cyndie Demeocq [11], Xiang Li [12], Arunav Das [13], Orlando Timmerman [14], Thomas Baldwin-McDonald [15], Jinge Wu [8], Peizhen Bai [2], Zahraa Al Sahili [16], Omnia Alwazzan [17], Thao N. Do [18], Mohammod N. I. Suvon [1], Angeline Wang [19], Lucia Cipolina-Kun [20], Luigi A. Moretti [21], Lucas Farndale [22], Nitisha Jain [13], Natalia Efremova [23], Yan Ge [12], Marta Varela [24], Hak-Keung Lam [9], Oya Celiktutan [9], Ben R. Evans [25], Alejandro Coca-Castro [3], Honghan Wu [26], Zahraa S. Abdallah [12], Chen Chen [2], Valentin Danchev [23], Nataliya Tkachenko [27], Lei Lu [28], Tingting Zhu [29], Gregory G. Slabaugh [17], Roger K. Moore [2], William K. Cheung [30], Peter H. Charlton [31] & Haiping Lu [2,32] ✉

[1]Centre for Machine Intelligence, University of Sheffield, Sheffield, UK. [2]School of Computer Science, University of Sheffield, Sheffield, UK. [3]The Alan Turing Institute, London, UK. [4]Department of Metabolism, Digestion and Reproduction, Imperial College London, London, UK. [5]Department of Applied AI, Simula Research Laboratory, Oslo, Norway. [6]Information School, University of Sheffield, Sheffield, UK. [7]NHS England, Leeds, UK. [8]Institute of Health Informatics, University College London, London, UK. [9]Department of Engineering, King's College London, London, UK. [10]Department of Computing Science, University of Aberdeen, Aberdeen, UK. [11]School of Informatics, University of Edinburgh, Edinburgh, UK. [12]School of Engineering Mathematics and Technology, University of Bristol, Bristol, UK. [13]Department of Informatics, King's College London, London, UK. [14]Department of Earth Sciences, University of Cambridge, Cambridge, UK. [15]Department of Computer Science, University of Manchester, Manchester, UK. [16]Department of Computer Science, Queen Mary University of London, London, UK. [17]Digital Environment Research Institute, Queen Mary University of London, London, UK. [18]Department of Computer Science, University of Bath, Bath, UK. [19]Department of Classics, King's College London, London, UK. [20]School of Electrical, Electronic and Mechanical Engineering, University of Bristol, Bristol, UK. [21]School of Engineering, University of the West of England, Bristol, UK. [22]Cancer Research UK Scotland Institute, Glasgow, UK. [23]School of Business and Management, Queen Mary University of London, London, UK. [24]City St George's University of London, London, UK. [25]British Antarctic Survey, Cambridge, UK. [26]School of Health and Wellbeing, University of Glasgow, Glasgow, UK. [27]Chief Data & AI Office, Lloyds Banking Group, London, UK. [28]School of Life Course & Population Sciences, King's College London, London, UK. [29]Institute of Biomedical Engineering, University of Oxford, Oxford, UK. [30]Department of Computer Science, Hong Kong Baptist University, Hong Kong, China. [31]Department of Public Health and Primary Care, University of Cambridge, Cambridge, UK. [32]These authors contributed equally: Xianyuan Liu, Jiayang Zhang, Shuo Zhou, Haiping Lu. ✉e-mail: h.lu@sheffield.ac.uk